

Lecture #7

Content to be covered

- ☐ Diffraction
- ☐ Classification of diffraction
- ☐ Superposition of N vibrations

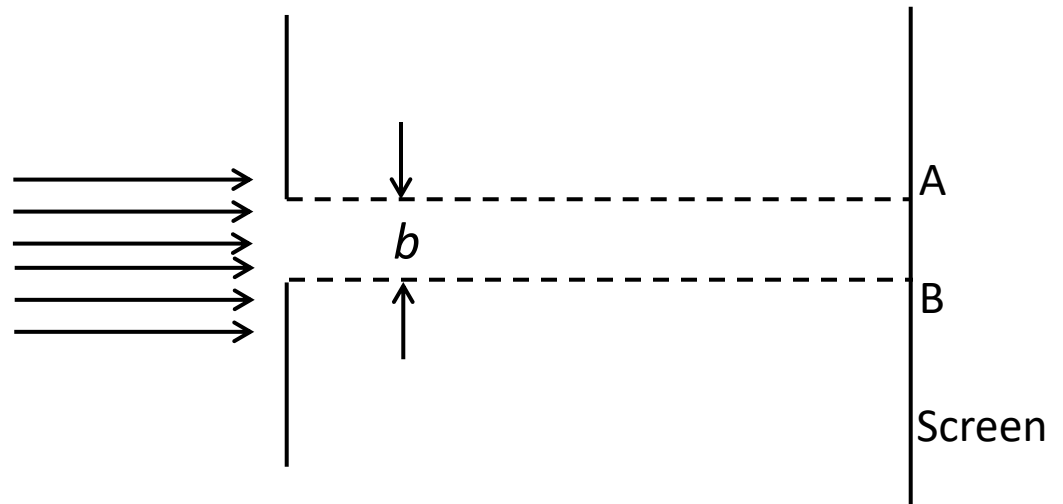
Resources to be consulted

- ☐ OPTICS by Ajay Ghatak

Diffraction

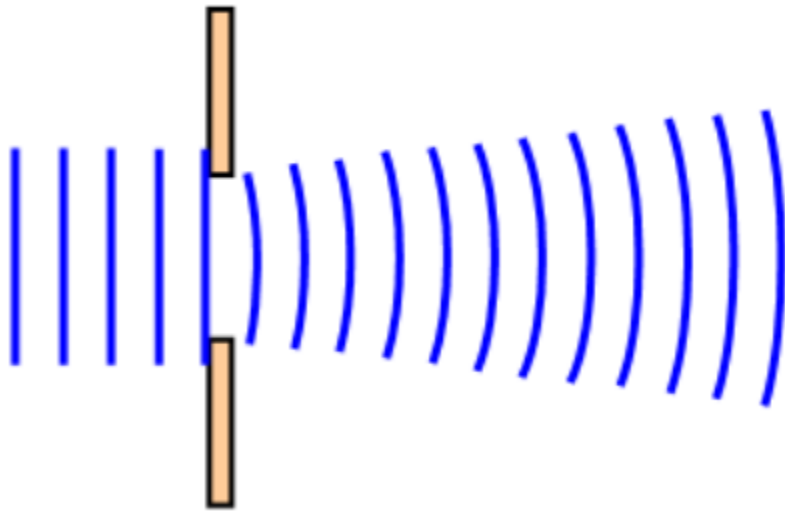
Diffraction of the light occurs when a light wave passes very close to the edge of an object or through a tiny opening, such as a slit or aperture.

According to geometrical optics region AB of screen to be illuminated and remaining portion will be dark.

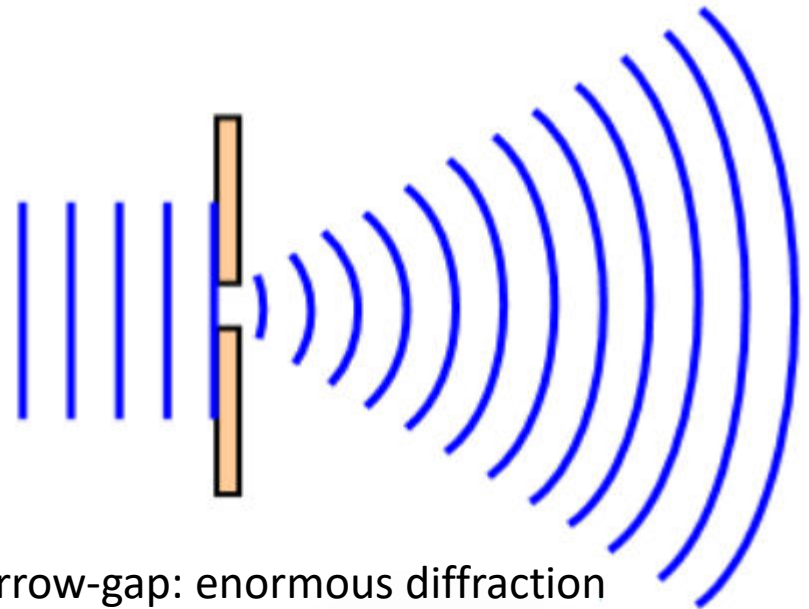


If observation are made carefully and width of the slit is not very large with respect to wavelength then the light intensity in the region AB is not uniform and there is also some intensity inside the geometrical shadow. **This spreading out of a wave when it passes through a narrow opening is known as diffraction pattern.**

Diffraction

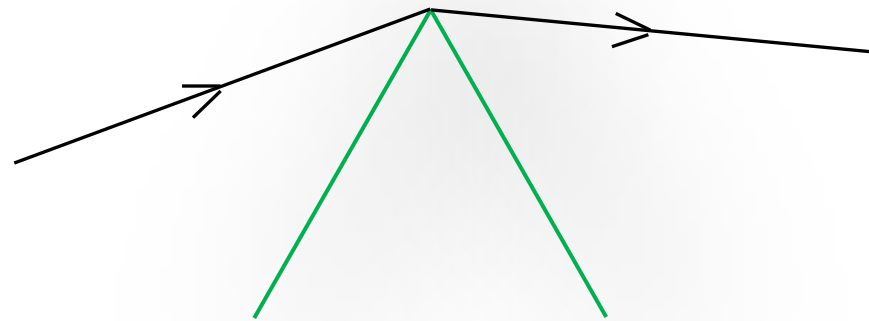


Wide-gap: slight diffraction

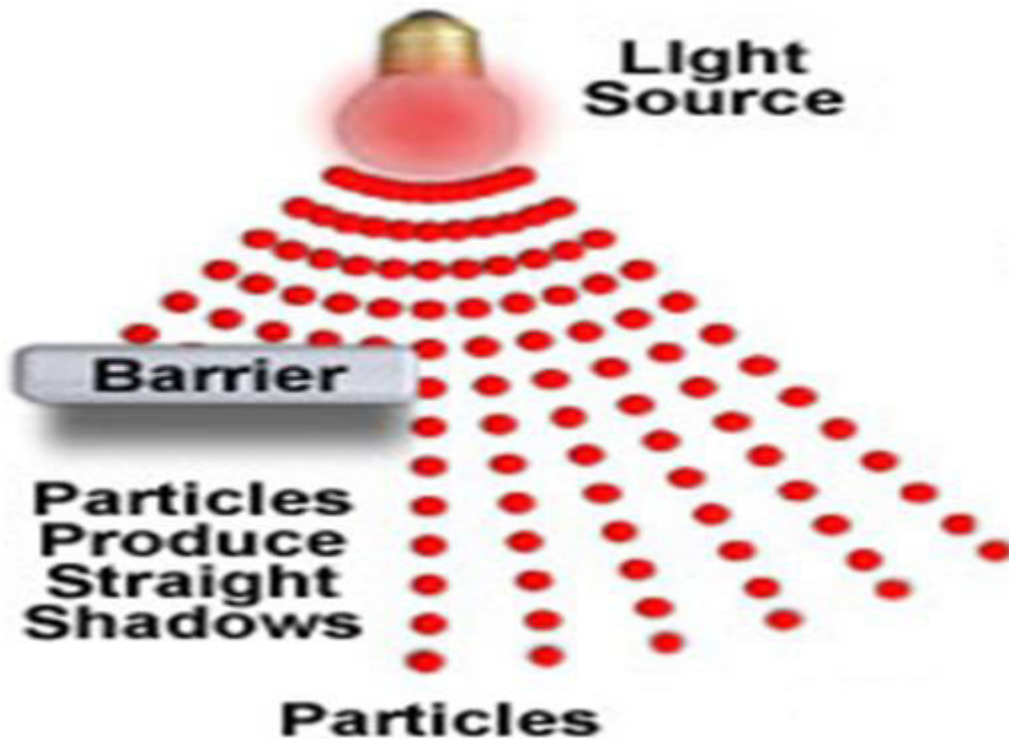


Narrow-gap: enormous diffraction

Particle theory does not explain this phenomena. On the basis of Huygen's principle, which state that all the points along a wave-front acts as they were a point source. So light coming out through the narrow opening will not act as rays rather than as a point source. Hence, they emerges out in all directions instead of just passing through slit as rays.



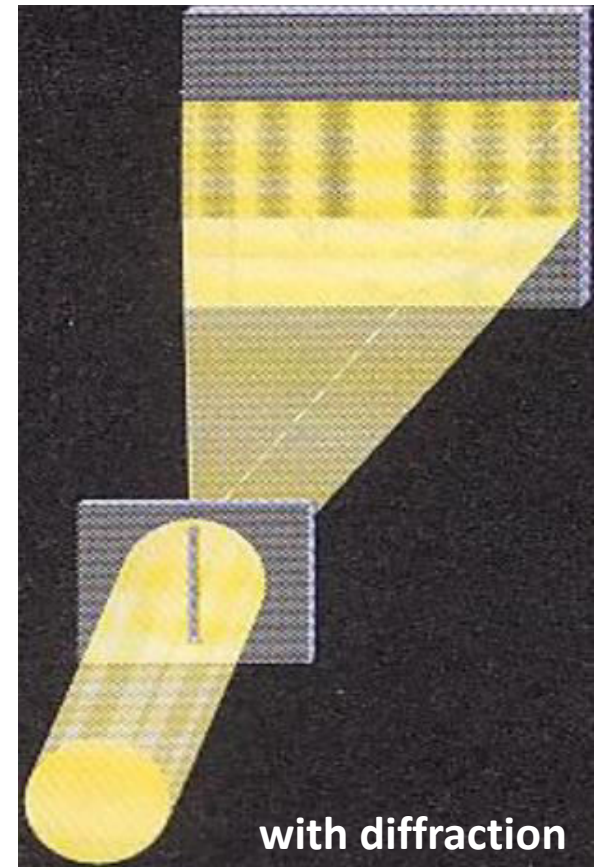
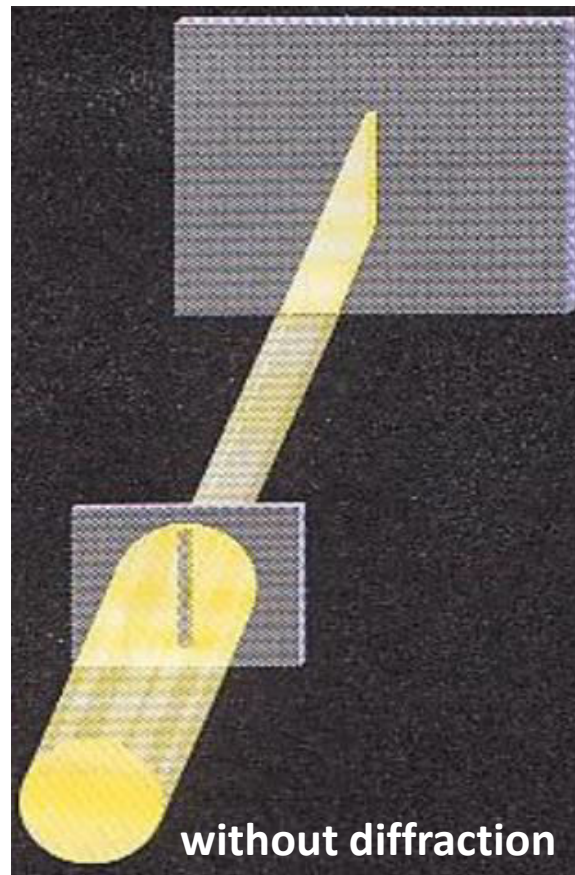
Diffraction of Particles and Waves



Definition of Diffraction

Bending of light round the corner of an obstacle or spreading out into the region of geometrical shadow when it passes through a narrow opening is known as the phenomenon of diffraction and the resulting distribution of light intensity on the screen is called diffraction pattern.

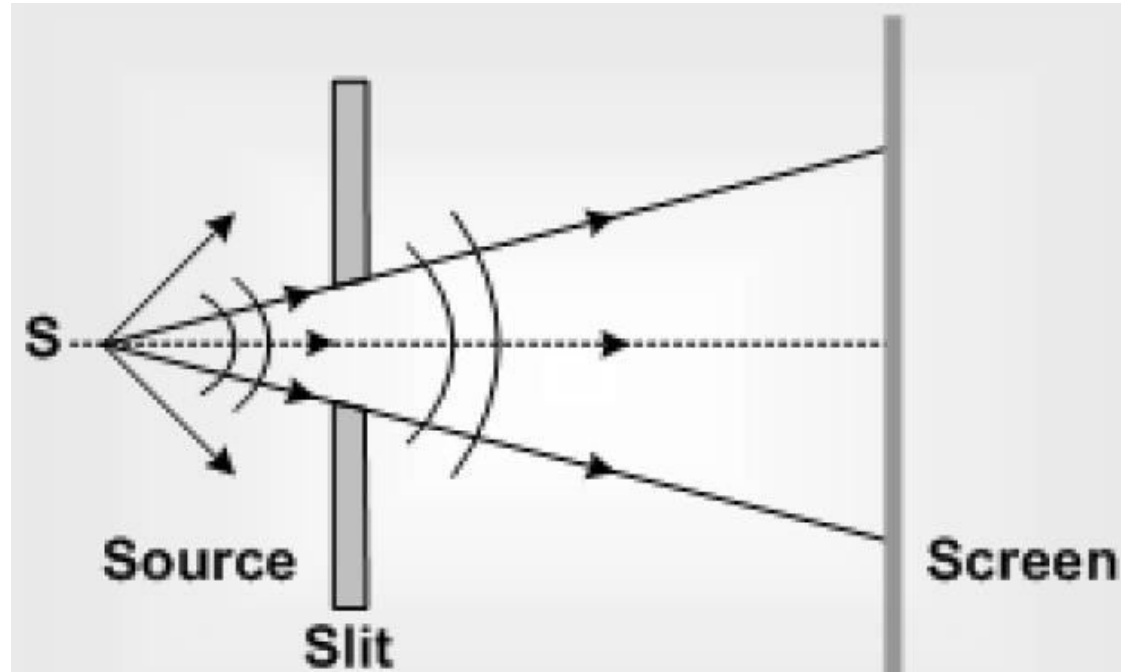
The spreading out of a wave when it passes through a narrow opening is known as diffraction pattern.



Types of Diffraction

Fresnel diffraction

The source of light and the screen are at finite distance from the diffracting aperture.

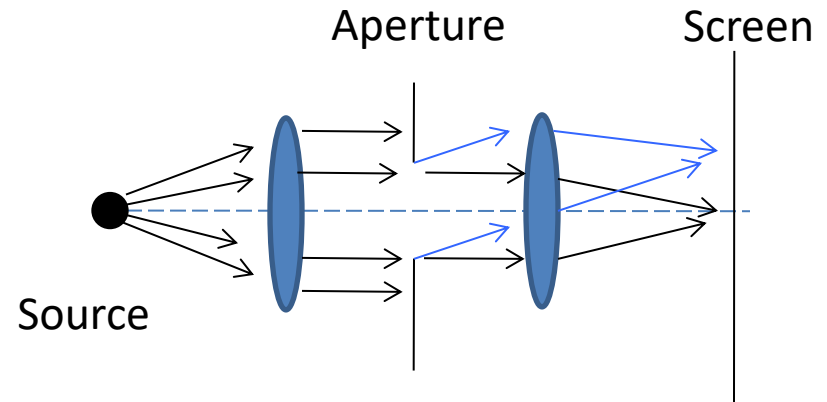


Wavelength of light must be comparable to the size of an obstacle.

Types of Diffraction

Fraunhofer diffraction

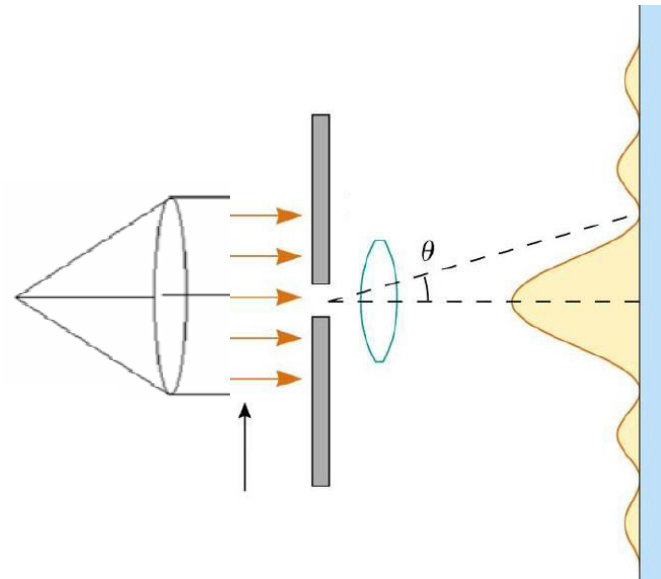
The source and the screen are at infinite distance from the aperture (Using convex lens)



The source is kept on the focal plane of 1st convex lens and the screen is also kept on the focal plane of other convex lens.

Incident & diffracted rays are parallel.

Both incident and diffracted waves may be considered to be plane wave.



Wavelength of light must be comparable to the size of an obstacle

Resultant of Simple Harmonic Waves of Equal Amplitude, Periods and Increasing Phases

Phases increasing in arithmetic progressions

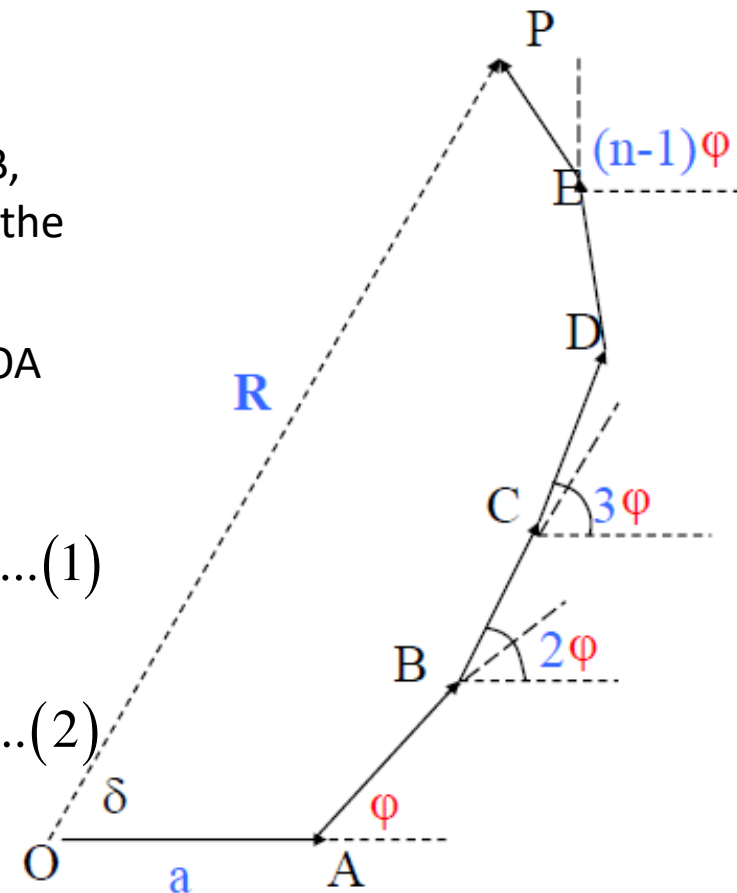
Let a be the amplitude of each vibration and ϕ the common phase difference between successive vibrations.

To find the amplitude and initial phase of the resultant vibration, we construct the polygon of amplitude $OA, AB, BC, \dots, P$, each equal to a . The closing side OP represent the resultant vibration.

Resolving the amplitude parallel and perpendicular to OA

$$R \cos \delta = a + a \cos \phi + a \cos 2\phi + a \cos 3\phi + \dots + a \cos(n-1)\phi \quad \dots\dots(1)$$

$$R \sin \delta = a \sin \phi + a \sin 2\phi + a \sin 3\phi + \dots + a \sin(n-1)\phi \quad \dots\dots(2)$$



$$2R \cos \delta \sin \frac{\varphi}{2} = 2a \sin \frac{\varphi}{2} + a \cos \varphi \sin \frac{\varphi}{2} + 2a \cos 2\varphi \sin \frac{\varphi}{2} + 2a \cos 3\varphi \sin \frac{\varphi}{2} + \dots$$

$$+ 2a \cos(n-1)\varphi \sin \frac{\varphi}{2}$$

Now using relation

$$2 \cos A \sin B = \sin(A+B) - \sin(A-B)$$

$$2R \cos \delta \sin \frac{\varphi}{2} = a \left[\begin{array}{l} 2 \sin \frac{\varphi}{2} + \sin \left(\varphi + \frac{\varphi}{2} \right) - \sin \left(\varphi - \frac{\varphi}{2} \right) + \\ \sin \left(2\varphi + \frac{\varphi}{2} \right) - \sin \left(2\varphi - \frac{\varphi}{2} \right) + \\ \sin \left(3\varphi + \frac{\varphi}{2} \right) - \sin \left(3\varphi - \frac{\varphi}{2} \right) + \\ \dots \dots \dots \dots \dots \dots \dots \dots \dots \dots \dots + \\ \sin \left((n-2)\varphi + \frac{\varphi}{2} \right) - \sin \left((n-2)\varphi - \frac{\varphi}{2} \right) + \\ \sin \left((n-1)\varphi + \frac{\varphi}{2} \right) - \sin \left((n-1)\varphi - \frac{\varphi}{2} \right) \end{array} \right]$$

$$2R \cos \delta \sin \frac{\varphi}{2} = a \sin \frac{\varphi}{2} + a \sin(n-1)\varphi = 2a \sin \frac{n\varphi}{2} \cos \frac{(n-1)\varphi}{2}$$

$$R \cos \delta = a \left(\frac{\sin \frac{n\varphi}{2}}{\sin \frac{\varphi}{2}} \right) \cos \frac{(n-1)\varphi}{2} \dots (3)$$

Similarly, we get

$$R \sin \delta = a \left(\frac{\sin \frac{n\varphi}{2}}{\sin \frac{\varphi}{2}} \right) \sin \frac{(n-1)\varphi}{2} \dots (4)$$

Now squaring and adding eq^{ns} (3) & (4)

$$R = a \frac{\sin \frac{n\varphi}{2}}{\sin \frac{\varphi}{2}}$$

Let $n\varphi = 2\alpha$ $R = a \frac{\sin \alpha}{\sin(\alpha/n)}$

Now suppose n is very large and amplitude a & phase φ are infinitely small

$$R = na \frac{\sin \alpha}{\alpha}$$

The quantity na is still finite (=A)

$$R = A \frac{\sin \alpha}{\alpha}$$

Now, dividing eq. (4) by eq. (3), we get

$$\tan \delta = \tan \frac{(n-1)\varphi}{2}$$

$$\Rightarrow \delta = \frac{(n-1)\varphi}{2}$$

$$\text{or } \delta \approx \frac{n\varphi}{2} = \alpha$$

If we have large number of vibration as $\cos(\omega t)$, $\cos(\omega t - \phi)$, $\cos(\omega t - 2\phi)$,.....
 $\cos[\omega t - (n-1)\phi]$; then resultant vibration will be

$$R \cos \delta = \frac{a \sin \frac{n\phi}{2}}{\sin \frac{\phi}{2}} \cos \left[\omega t - \frac{(n-1)\phi}{2} \right]$$

Resultant intensity at P which is proportional to the square of the resultant amplitude R is given by:

$$I = R^2 = A^2 \frac{\sin^2 \alpha}{\alpha^2} = I_0 \frac{\sin^2 \alpha}{\alpha^2}$$

Resultant wave will be given by

$$E = a [\cos \omega t + \cos(\omega t - \phi) + \dots + \cos(\omega t - (n-1)\phi)]$$

$$E = R \cos \left[\omega t - \frac{(n-1)\phi}{2} \right]$$